

Riemannian Geometry

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1 Smooth Manifolds

1.1 Simple pre-requisites

This chapter will heavily be built on the basics of topology. This section will be a recap of some important properties.

Definition 1.1 (A topology on a set). Given a set X , we define τ to be a collection of subsets from X that follow certain rules:

1. The empty set $\emptyset$ and X are in τ .
2. If we have a collection of sets in τ , then their union is also in τ . i.e. If $U_\alpha \in \tau \forall \alpha \in I$, then $\bigcup_{\alpha \in I} U_\alpha \in \tau$
3. If $U_1, \dots, U_n \in \tau$ then $\bigcap_{i=1}^n U_i \in \tau$

We call the elements in τ open sets and we use this along with our set X to create a topological space.

Definition 1.2 (A topological space). A topological space is the pair (X, τ_X) where X is just a set and τ_X is the topology on X

Instead of always writing (X, τ_X) , we use X to refer to the space.

Definition 1.3 (A continuous function between topological spaces). Given 2 topological spaces X, Y , a function $f : X \rightarrow Y$ is said to be continuous if for every open set $V \subseteq Y$, the preimage $f^{-1}(V) = \{x \in X \mid f(x) \in V\}$ is open in X .

This definition does not depend on a metric, so it allows us to define continuity on general topological spaces.

Definition 1.4 (Homeomorphisms). Given 2 topological spaces X and Y . A function $f : X \rightarrow Y$ is called a homeomorphism if:

1. f is bijective
2. f is continuous
3. f^{-1} is continuous

If 2 spaces X and Y have a homeomorphism between them, we call them homeomorphic and use $X \cong Y$ to represent that.

This notion of topology allows us to define spaces that locally resemble Euclidean space without requiring a global coordinate system.

1.2 Topological Manifolds

Definition 1.5 (Topological Manifold). Let M be a topological space. M is a **topological n -manifold** (topological manifold of dimension n) if it is:

- A Hausdorff space: For any 2 distinct points $p, q \in M$, there exists 2 disjoint open subsets $U, V \subseteq M$ such that $p \in U$ and $q \in V$.
- Second countable: there exists a countable basis for the topology of M
- M is locally Euclidean of dimension n : each point $p \in M$ has a neighbourhood that is homeomorphic to an open subset of $\mathbb{R}^n$

A very simple example of a topological manifold is the space $\mathbb{R}^n$ itself. We'll cover more interesting spaces later.

Definition 1.6 (Coordinate Charts). Let M be a topological n -manifold. A coordinate chart on M is a pair (U, φ) where U is an open subset of M and φ is a homeomorphism $f : U \rightarrow \hat{U}$ where $\hat{U} \subseteq \mathbb{R}^n$. We say the chart is centred at p if $\varphi(p) = 0$

Charts are very useful, as without them, calculus on these manifolds would be next to impossible. Coordinate charts give us a way of easily moving from the complex, strange manifolds to simple Euclidean space. Now that we have these definitions, we can start giving examples of more interesting manifolds.

Example 1.1 (Spheres). For any $n \geq 0$, the n -sphere is defined as:

$$\mathbb{S}^n = \left\{ (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid \sum_{i=1}^{n+1} x_i^2 = 1 \right\}$$

If we equip this from the subspace topology inherited from $\mathbb{R}^{n+1}$, This forms a topological n -manifold. Since $\mathbb{R}^{n+1}$ is a metric space, it is Hausdorff and second countable. Because $\mathbb{S}^n$ uses this subspace topology, it also inherits them.

To show that it is locally Euclidean, we must show that every point $p \in \mathbb{S}^n$ has a neighbourhood that is part of a coordinate chart (U, φ) .

Let $N = (0, \dots, 0, 1)$ denote the North pole and $S = (0, \dots, 0, -1)$ denote the south pole. We define the 2 open sets of $\mathbb{S}^n$ as:

$$U_N = \mathbb{S}^n \setminus \{N\}, \quad U_S = \mathbb{S}^n \setminus \{S\}$$

First consider $p \in \mathbb{S}^n$ such that $p \neq N$. Then $p \in U_N$. We can now define the stereographic projection map $\varphi_N : U_N \rightarrow \mathbb{R}^n$ as:

$$\varphi_N(x_1, \dots, x_{n+1}) = \left(\frac{x_1}{1 - x_{n+1}}, \dots, \frac{x_n}{1 - x_{n+1}} \right)$$

This is clearly a homeomorphism as it is bijective and continuous. Its inverse is also continuous; however, due to limitations (i'm lazy), it will not be shown

in this document. Since φ_N is a homeomorphism between U_N and $\mathbb{R}^n$, the pair (U_N, φ_N) is a coordinate chart containing p . Now consider the point $p = N$. This would mean $p \in U_S$. We can define the projection from the south pole as:

$$\varphi_S(x_1, \dots, x_{n+1}) = \left(\frac{x_1}{1 + x_{n+1}}, \dots, \frac{x_n}{1 + x_{n+1}} \right)$$

Since it very similar to φ_N , we can use similar reasoning/proof methods to show it is also a homeomorphism. It also means that (U_S, φ_S) is also a valid coordinate chart containing N . Since every point is contained in at least one of these charts. $\mathbb{S}^n$ is locally Euclidean of dimension n . We have now officially shown that $\mathbb{S}^n$ is a topological n -manifold.

Example 1.2 (Product Manifolds). Suppose $M_1, \dots, M_k$ are topological manifolds of dimensions $n_1, \dots, n_k$ respectively. The product space $M_1 \times \dots \times M_k$ is a topological manifold of dimension $n_1 + \dots + n_k$ as follows. It is obviously Hausdorff and second countable, so the only property that needs to be checked is locally Euclidean. Given any point $(p_1, \dots, p_k) \in M_1 \times \dots \times M_k$, we can choose the coordinate chart (U_i, φ_i) for each M_i with $p_i \in U_i$. The product map:

$$\varphi_1 \times \dots \times \varphi_k : U_1 \times \dots \times U_k \rightarrow \mathbb{R}^{n_1 + \dots + n_k}$$

is a homeomorphism onto its image, which is just a product open subset of $\mathbb{R}^{n_1 + \dots + n_k}$. Thus $M_1 \times \dots \times M_k$ is a topological manifold of dimension $n_1 + \dots + n_k$.

Example 1.3 (Tori). for $n \in \mathbb{N}$, the n -torus is the product space: $\mathbb{T}^n = \mathbb{S}_1^1 \times \dots \times \mathbb{S}_n^1$. Since we proved that product manifolds are manifolds, it follows pretty simply that the n -torus is a manifold as well.

Before we finish this section, one thing we need to introduce is the notion of an Atlas.

Definition 1.7 (Atlas). Given a manifold M , an atlas is a collection of coordinate charts $\mathcal{A} = \{(U_\alpha, \varphi_\alpha) : \alpha \in I\}$ such that: $M = \bigcup_{\alpha \in I} U_\alpha$